

MIN-MAX ± 1 QUADRATIC FORM

MathOverflow 413935 · existence of the limit

THEOREM

$$L = \lim_{n \rightarrow \infty} \alpha_n = \frac{1}{2}$$

The limit exists and equals $\frac{1}{2}$.

THE QUESTION

$$L = \lim_{n \rightarrow \infty} n^{-3/2} \min_{a_{ij} = \pm 1} \max_{x_i = \pm 1} \left| \sum_{i < j} a_{ij} x_i x_j \right|$$

Write $\alpha_n = m_n/n^{3/2}$. Does $\lim \alpha_n$ exist?

ARGUMENT

1. Sandwich: $1/\pi \leq \liminf \alpha_n \leq \limsup \alpha_n \leq 1/2$.
2. On the dense Paley family $n = p^2 + 1$, $\Phi(C) = \frac{1}{2}n\sqrt{n-1}$.
If $m_n \geq \Phi(C) - O(1)$, denseness forces $L = \frac{1}{2}$.
3. Every gap-2 undercutter of C is empty or non-descending;
hence $m_n \geq \Phi(C) - 2$ and $L = \frac{1}{2}$.

REMARK

An independent residual / 16N spectral package remains open;
it is not required for the limit.

OUTLINE OF THE PROOF

bi-tight emptiness · Type I · deep freeness-fail · denseness

① BI-TIGHT COVERS

No size- $2p$ bi-tight $\text{Max}^\pm$ cover of the Paley conference exists for primes $p \geq 5$.

Multiplicity of $\lambda_{\max}$ and a lower bound on $\lambda_{\min}$ force the cycle

eigenvalue below the bi-tight threshold (majorization; residual / $16N$ unused).

② TYPE-I FREENESS FAILURE

The remaining Type-I gap-2 candidate ($k = 3p - 2$) dualizes to a linear system

on edge variables. Box and sum constraints are contradictory for every $p \geq 5$.

Hence that class admits no further descent.

③ DEEP FREENESS FAILURE

Deep covers ($k \geq 3p$) are either free, fail-equality empty, or dualize to a

two-level system obstructed by the same box lower bound.

With freeness and tight emptiness: $m_n \geq \Phi(C) - 2$.

CONCLUSION

$L = \frac{1}{2}$. Writeup and verification: github.com/luckyseoul/quadratic-minmax-limit